

WHEN INLIER RANKING FAILS TO CONTROL GALLERY INFLUENCE IN FROZEN FEW-SHOT CLASSIFICATION

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ABSTRACT

Unlabeled galleries may contain many classes outside a few-shot episode. We examine a gallery-only transfer of established open-set prototype fitting with frozen visual features and queries excluded from adaptation. In two image-disjoint Oxford-IIIT Pets pools, source-default matched one-shot accuracy is 94.92% and 94.87%, below retrieval-weighted controls at 96.24% and 96.02%. Mean inlier ranking AUROC is approximately 0.93, yet about 80% of retained gallery weight is task-external. Retrospective final-step mass balancing removes most mismatched-gallery harm but leaves a matched one-shot deficit. Oracle composition filtering produces a separate diagnostic gain using unavailable inference labels. The results distinguish ranking quality, retained mass and composition within a fixed transfer protocol. They establish neither a new algorithm nor a failure of original transductive OSLO or independent-domain generalization.

1 INTRODUCTION

Independent unlabeled galleries are attractive resources for few-shot classification, but a gallery can contain mostly classes outside a given episode. Small relevance weights on individual samples need not imply a small accumulated contribution from irrelevant samples.

We examine this distinction in a restricted gallery-only transfer of Open-Set Likelihood Optimization (OSLO; Boudiaf et al., 2023). Prototypes are fitted from labeled support and an independent gallery; queries do not participate in adaptation. The source-default transfer underperforms strong local controls in every evaluated condition, despite favorable inlier ranking. The contribution is a scoped empirical diagnosis of that discrepancy, not an improved classification algorithm.

Three results structure the analysis: source-default fitting improves on a closed-set ablation but fails against stronger classifiers; approximately four-fifths of retained gallery weight is task-external despite mean inlier AUROC near 0.93; and final-step mass and composition interventions have different effects. The conclusions concern a fixed transfer on Oxford-IIIT Pets, not original transductive OSLO or a general impossibility result.

1.1 RELATED WORK

Distractor-containing unlabeled episodes and prototype augmentation are established in semi-supervised few-shot learning (Ren et al., 2018). OSLO alternates latent inlier scores, class assignments and parametric fitting in a transductive open-set problem (Boudiaf et al., 2023). Our study changes the unlabeled pool from queries to an independent gallery and evaluates closed-set query classification.

Partial optimal transport has also been used to derive an outlier mass score and train an OOD detector jointly with a semi-supervised classifier (Ren et al., 2024). Transported mass in that learned selection objective differs from the scalar averaging coefficients inspected here. Our retrospective interventions characterize a fixed gallery-only predictor and do not establish an alternative to learned open-set selection.

Simple feature transformations can materially affect few-shot classification (Wang et al., 2019), motivating centering and strong-classifier controls. Retrieval augmentation can also include encoder

finetuning, as in SWAT (Liu et al., 2024), which differs from the frozen-feature regime here. We make no numerical comparison to the original OSLO or SWAT benchmark tables. The objectives, encoders and preprocessing operations used here are existing methods.

2 METHOD: A GALLERY-ONLY TRANSFER

An episode contains $C = 5$ classes, k labeled examples per class, and a fixed gallery of N unlabeled features. CLIP and DINOv2 embeddings (Radford et al., 2021; Oquab et al., 2023) are normalized and fused with equal contribution without updating either encoder. Support and gallery jointly determine a centering vector, followed by feature normalization. Queries do not contribute to these operations. Let s_{ci} and g_j denote transformed support and gallery features.

Initial prototypes are $\mu_c^{(0)} = k^{-1} \sum_i s_{ci}$ and class assignments start uniformly. In each of two source-default updates, cosine logits ℓ_{jc} are computed against normalized current prototypes, followed by

$$\xi_j = \sigma \left(\lambda_s^{-1} \sum_c z_{jc} \ell_{jc} \right), \quad z_{jc} = \text{softmax}_c(\xi_j \ell_{jc} / \lambda_z), \quad (1)$$

$$w_{jc} = \xi_j z_{jc}, \quad \mu_c = \frac{k \mu_c^{(0)} + \sum_j w_{jc} g_j}{k + \sum_j w_{jc}}, \quad (2)$$

where $\lambda_s = 0.05$, $\lambda_z = 0.1$, and the inlier update uses the preceding assignments. Each query is centered with the fitted vector, normalized and classified by cosine similarity to final prototypes.

The closed-set ablation replaces ξ_j by one in the assignment and prototype updates. Zero update retains support-and-gallery centering but predicts with initial support prototypes; it is not a wholly gallery-free baseline. Gallery labels are absent from eligible inference interfaces.

2.1 WEIGHT ACCOUNTING AND INTERVENTIONS

Let $M_c = \sum_j w_{jc}$ be final gallery mass and $\rho_c = M_c / (k + M_c)$ its fraction of the unnormalized support-plus-gallery average. We also inspect the fraction of gallery weight on classes outside the episode. True labels enter only this retrospective accounting. Feature directions and final cosine normalization prevent treating these scalar coefficients as causal importance scores.

Mass balancing replaces nonzero weights by $k w_{jc} / M_c$, matching total gallery mass to support mass. Oracle composition removes examples whose true class is outside the episode, retaining the earlier fitted weights for all remaining examples. It does not remove wrong assignments among episode classes. A combined intervention filters then balances; zero remaining mass falls back to the support prototype. Centering and all preceding updates stay fixed. These interventions were defined after the main result and are explanatory diagnostics, not prospectively qualified predictors.

3 EXPERIMENTS

3.1 PROTOCOL

Oxford-IIIT Pets contains 37 breeds (Parkhi et al., 2012). We evaluate an original fixed image pool and 1,804 previously unused confirmation images with zero exact decoded-RGB overlap with the original query and gallery pools. Confirmation shares the original classes and domain. Source defaults and inference code were frozen before confirmation feature extraction; original outcomes had already been inspected. Both pools became exposed by subsequent diagnostics.

For each pool and support count, five seeds generate 100 episodes each. Episodes contain five classes and 15 queries per class, with disjoint support and query identities. Each episode is evaluated with a 1,876-image Pets gallery and a 1,876-image DTD gallery (Cimpoi et al., 2013): 2,000 sampled episodes produce 4,000 task-gallery-condition evaluations. Queries are always Pets images. The independent galleries are disjoint from support/query pools; matched denotes the target domain, not a restriction to the five episode classes.

Pool / gallery / shot	OSLO _G	Closed	Zero	R2	CS _{ℓ₂}	LR1	LR10
Original / Pets / 1	94.92	90.50	94.75	96.24	96.26	94.28	94.33
Original / DTD / 1	64.19	36.93	94.40	93.93	93.96	94.28	94.33
Original / Pets / 5	97.19	94.38	98.44	98.44	98.55	98.41	98.56
Original / DTD / 5	93.30	49.41	98.46	98.44	98.55	98.41	98.56
Confirm. / Pets / 1	94.87	91.70	94.22	96.02	96.05	93.73	93.74
Confirm. / DTD / 1	63.99	38.08	93.74	93.39	93.44	93.73	93.74
Confirm. / Pets / 5	97.19	94.70	98.38	98.35	98.31	98.34	98.38
Confirm. / DTD / 5	92.98	48.76	98.35	98.35	98.31	98.34	98.38

Table 1: Accuracy (percent) for all conditions, with 500 paired tasks per cell. Confirmation is an image-disjoint pool from the same target dataset. Queries are always Pets images; DTD is the mismatched gallery. Closed removes inlier weighting; Zero retains initial centered support prototypes. LR1 and LR10 are support-only logistic controls with fixed regularization.

All seven methods use frozen CLIP ViT-B/16 and DINOv2 ViT-S/14 features. Controls are the local retrieval-weighted classifier R2, its support-centered control CS_{ℓ₂}, support-only multinomial logistic classifiers at fixed $C = 1$ and $C = 10$, and the two source ablations. R2 and CS_{ℓ₂} denote frozen local comparison implementations, not new methods. Queries cannot update features, relevance weights or prototypes; gallery labels cannot enter inference.

Paired accuracy intervals use 5,000 bootstrap resamples stratified within each pool’s five seeds. Gallery conditions share resampling indices. Intervals condition on fixed image pools and quantify task variation, not independent image or domain sampling. Pointwise comparisons are descriptive without a global multiplicity claim.

3.2 STRONG CONTROLS CHANGE THE INTERPRETATION

OSLO_G improves matched one-shot accuracy over closed-set fitting by 4.413 and 3.171 percentage points in the original and confirmation pools. That component effect does not establish system superiority. R2 achieves 96.240% and 96.016%, versus 94.917% and 94.875% for OSLO_G. Paired differences are -1.323 pp (95% interval $[-1.723, -0.933]$) and -1.141 pp ($[-1.517, -0.763]$).

With a DTD gallery, one-shot OSLO_G accuracy falls to 64.192% and 63.992%, versus R2 at 93.931% and 93.389%. All five-shot conditions also degrade. All eight paired intervals against R2 are negative (Appendix A). Every breed has a negative descriptive difference in each mismatched one-shot pool; this is not 37 independent significance tests.

Zero update separates update effects from preprocessing. Its original matched one-shot comparison with OSLO_G has a zero-containing interval, whereas OSLO_G exceeds it on confirmation matched one-shot tasks. Zero update avoids most mismatch harm. Thus the fixed defaults fail strong-control qualification even though latent inlier weighting improves the closed-set update.

4 ANALYSIS

4.1 RANKING AND ACCUMULATED WEIGHT DIVERGE

Matched one-shot mean task AUROC is 0.9280 and 0.9263, but 80.408% and 80.826% of retained gallery weight is task-external. Ranking and retained mass answer different questions: the latter also depends on the number of irrelevant samples and their absolute weights. Good ordering does not demonstrate calibrated membership probabilities.

In mismatched one-shot conditions every gallery example is task-external, yet the final gallery coefficient averages approximately 99.5%. Little scalar mass remains on support in the final unnormalized average. This is an algebraic observation, not a measure of feature-vector direction or a causal contribution for each image. All 4,000 task-condition predictions replay exactly, linking the accounting to the evaluated procedure.

Pool / gallery / shot	Gallery coeff. (%)	External share (%)	Mean AUROC
Original / Pets / 1	99.597	80.408	0.9280
Original / DTD / 1	99.485	100.000	–
Original / Pets / 5	98.053	80.731	0.9477
Original / DTD / 5	97.424	100.000	–
Confirm. / Pets / 1	99.605	80.826	0.9263
Confirm. / DTD / 1	99.487	100.000	–
Confirm. / Pets / 5	98.071	80.862	0.9510
Confirm. / DTD / 5	97.432	100.000	–

Table 2: Final-update weight accounting. Gallery coefficient is the mean class-specific gallery fraction of support-plus-gallery mass. External share is the fraction of gallery weight on classes outside the episode. AUROC is undefined if no in-episode class is present. These descriptive coefficients are not causal directional effects.

Pool / gallery / shot	Original	Mass	Oracle	Oracle+mass	Zero	R2
Original / Pets / 1	94.92	95.76	97.92	96.94	94.75	96.24
Original / DTD / 1	64.19	94.30	94.40	94.40	94.40	93.93
Original / Pets / 5	97.19	98.59	98.79	98.83	98.44	98.44
Original / DTD / 5	93.30	98.42	98.46	98.46	98.46	98.44
Confirm. / Pets / 1	94.87	95.39	97.87	96.74	94.22	96.02
Confirm. / DTD / 1	63.99	93.61	93.74	93.74	93.74	93.39
Confirm. / Pets / 5	97.19	98.39	98.58	98.61	98.38	98.35
Confirm. / DTD / 5	92.98	98.32	98.35	98.35	98.35	98.35

Table 3: Retrospective final-step interventions, accuracy in percent. Mass matches gallery weight to support mass per class. Oracle removes examples whose true class is outside the episode; Oracle+mass combines both. Oracle variants require privileged labels. Earlier fitted weights and centering stay fixed.

4.2 MASS AND COMPOSITION HAVE DISTINCT EFFECTS

Final mass balancing raises mismatched one-shot accuracy from 64.192% to 94.301% in the original pool and from 63.992% to 93.613% in confirmation. Most mismatch harm can be removed by this constrained intervention. However, matched one-shot accuracy remains below R2 by 0.483 and 0.624 pp, with both paired intervals negative (Appendix B). Mass correction alone does not meet the improvement objective.

Oracle filtering gives matched one-shot accuracies of 97.923% and 97.867%, exceeding R2 by 1.683 and 1.851 pp. This diagnoses composition in an already-fitted state; it is not an admissible deployment comparison because gallery labels are unavailable at inference. It is also not a theoretical upper bound: earlier contaminated updates remain fixed. Combining filtering and balancing changes performance again, so the two operations are not interchangeable.

Both diagnostics reuse the main evaluation tasks rather than adding independent test samples. Weight accounting is descriptive; final-step interventions isolate particular changes to a saved fitted state. Neither establishes that a learned membership estimator will generalize or that membership is the only useful next design direction.

5 LIMITATIONS

The primary experiment covers one target dataset, one frozen feature fusion, two fixed galleries and source-default parameters. Distinct confirmation images add within-domain evidence, not new classes or a domain. The large gallery may itself contribute to accumulated mass; gallery-size and hyperparameter sensitivity remain unmeasured.

The protocol differs from original OSLO and does not reproduce its benchmark tables. A closed-set ablation cannot replace strong competitive baselines. The frozen local controls have explicit protocol boundaries, and no comprehensive state-of-the-art ranking is established.

Interventions are retrospective and oracle filtering uses privileged labels. Exact decoded-image duplicates are excluded, but near-duplicates and overlap with encoder pretraining data are not audited. Bootstrap uncertainty is conditional on fixed pools; no independent-domain confidence or family-wise significance claim is made. No matched per-method latency or energy experiment is available, so no efficiency advantage is asserted.

A separate source-trained calibration experiment on two previously used development domains is reported in Appendix E. Its two-direction accuracy gate fails and its predictions match a same-mass control. It provides a limited additional failure boundary, not independent confirmation of a remedy.

6 CONCLUSION

In the evaluated gallery-only transfer, favorable inlier ranking does not prevent harmful accumulated weight from task-external classes. Strong controls expose the accuracy deficit, and mass and composition interventions distinguish two contributors without qualifying a new method. Aggregate retained weight should be assessed alongside relevance ranking and downstream accuracy under explicit information permissions.

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Pool / gallery / shot	Δ vs R2 (pp)	Paired 95% interval
Original / Pets / 1	-1.323	[-1.723, -0.933]
Original / DTD / 1	-29.739	[-30.744, -28.792]
Original / Pets / 5	-1.248	[-1.552, -0.960]
Original / DTD / 5	-5.139	[-5.573, -4.728]
Confirm. / Pets / 1	-1.141	[-1.517, -0.763]
Confirm. / DTD / 1	-29.397	[-30.341, -28.456]
Confirm. / Pets / 5	-1.165	[-1.472, -0.877]
Confirm. / DTD / 5	-5.376	[-5.781, -4.968]

Table 4: Source-default transfer differences. Intervals use 5,000 paired bootstrap resamples stratified within task seeds and condition on fixed pools, not independent domains.

Condition / intervention	Δ vs R2 (pp)	Paired 95% interval
Original / Pets / 1 / Mass	-0.483	[-0.683, -0.293]
Original / Pets / 1 / Oracle	+1.683	[+1.397, +1.973]
Original / Pets / 1 / Oracle+mass	+0.699	[+0.544, +0.853]
Confirm. / Pets / 1 / Mass	-0.624	[-0.784, -0.456]
Confirm. / Pets / 1 / Oracle	+1.851	[+1.571, +2.144]
Confirm. / Pets / 1 / Oracle+mass	+0.728	[+0.560, +0.912]

Table 5: Matched one-shot diagnostic effects against R2. Intervals are conditional and retrospective; oracle gains do not qualify an admissible predictor.

A PAIRED PERFORMANCE DIFFERENCES

Original task seeds are 26091311–26091315 and confirmation seeds are 26091331–26091335, with 100 episodes per seed and support count. Each episode is reused across gallery conditions. Bootstrap seed is 26091399. Complete per-seed and breed values accompany the retained source data as descriptive breakdowns.

B PAIRED DIAGNOSTIC DIFFERENCES

Both diagnostics reproduce original predictions exactly. With a fully mismatched gallery, oracle filtering eliminates all gallery mass and agrees with zero update. Balanced nonempty galleries have scalar coefficient one-half. These checks validate the finite intervention, not an independent algorithm qualification.

C FROZEN COMPARISON IMPLEMENTATIONS

R2 first normalizes each encoder block and concatenates them with equal squared-norm contribution. For one-shot tasks it retrieves the 64 gallery features most similar to each normalized support prototype, normalizes their mean, and averages that vector with the prototype at equal weight before normalizing again. A five-class ridge classifier is fitted to these five augmented representatives and one-hot targets, using regularization $\lambda = 0.1$. For five-shot tasks it instead fits the original 25 support features with $\lambda = 1$; its predictions then do not depend on the gallery.

CS_{ℓ_2} subtracts the support feature mean from support, query and eligible gallery features and normalizes each resulting vector. It uses the same one-shot retrieval and augmentation recipe in the transformed geometry, and a ridge penalty of 0.1 for both support counts. Thus this frozen control differs from R2 in centering and, for five-shot tasks, regularization; their contrast is not a pure centering ablation.

Both ridge heads use an unpenalized intercept: with row-centered training features $A_c = A - \bar{A}$ and targets $Y_c = Y - \bar{Y}$, scores are

$$(Q - \bar{A})A_c^\top (A_c A_c^\top + \lambda I)^{-1} Y_c + \bar{Y}.$$

LR1 and LR10 fit support-only multinomial logistic regression on the same uncentered fused representation, using the L-BFGS solver, $C = 1$ or 10 , tolerance 10^{-8} and at most 2,000 iterations. No convergence warning occurred in the retained evaluations. These controls use fixed settings rather than parameter selection on confirmation queries.

D REPRODUCIBILITY AND SOURCE AVAILABILITY

Retained evaluation materials include the fixed protocol, code/model hashes, image-identity manifests, task indices, predictions and reference-implementation checks. Canonical comparator predictions match the frozen outputs exactly. Source parity and query permutation, splitting and insertion checks are retained separately from the complete task-identity and accuracy audit.

The prototype-fitting source is pinned to public OSLO revision `9240a4e630874c50459e069db609db9163a81c03`. The Method section specifies its restriction: support and independent gallery determine centering and fitting; queries only receive predictions. Feature extraction used float32 on an integrated GPU, with normalized half-precision feature storage. Evaluation and bootstrap environments are retained in manifests.

Public dataset and original model/code sources are cited above. Derived feature caches and task outputs have not been deposited publicly. No public reproducibility package or dataset accession is claimed; the review materials preserve provenance for subsequent checking.

E SOURCE CALIBRATION UNDER DOMAIN SHIFT

This separate development experiment tests a possible remedy, rather than extending the primary OSLO comparison. All DTD and EuroSAT pools had been used in earlier development; these outcomes provide no prospective confirmation on an unexposed dataset. DTD (Cimpoi et al., 2013) and EuroSAT (Helber et al., 2017) use the same frozen feature fusion as the primary evaluation. A model is trained on each domain and evaluated on the other, with matched and fully mismatched galleries.

Source development and inference permissions. Each source uses 100 one-shot episodes to train a binary classifier of gallery membership in the episode’s classes. Supports come from one half of the available source query pool. Another 100 episodes from the other half select one fixed control using source-domain accuracy and a matched source gallery. Training supports and control-selection images are disjoint. Source gallery labels are used only for training; target gallery labels and query labels are excluded from adaptation. The same source-selection budget is available to the listed controls, and the calibrated model’s architecture and penalty are fixed.

Seven statistics describe a gallery sample relative to the support prototypes: maximum cosine similarity and the gap between its two highest similarities for each encoder and the fused features, plus agreement between the encoders’ highest-scoring classes. Standardized statistics feed a class-balanced logistic regressor with $C = 1$, L-BFGS tolerance 10^{-8} and at most 1,000 iterations. Its output p is treated as a posterior under reference prior one-half. This is a fixed approximation, not a demonstrated guarantee of probability calibration.

Prior correction and controls. Maximum-likelihood prior adjustment is established under label-shift assumptions (Alexandari et al., 2020). Here the binary corrected posterior is

$$q_\pi(p) = \frac{\pi p}{\pi p + (1 - \pi)(1 - p)}.$$

Starting from $\pi = 0.5$, each iteration sets π to the gallery average of $q_\pi(p)$, stopping at change below 10^{-7} or 200 iterations; input probabilities are clipped to $[10^{-6}, 1 - 10^{-6}]$. Transfer of class-conditional score distributions is an assumption to test, not an asserted property of these domains.

For one-shot classification, the 64 neighbors retrieved by R2 are averaged using these posterior weights and normalized. The mixture coefficient with the support prototype is one-half of their mean posterior. The ridge penalty is 0.1. The same-mass control uses exactly that coefficient with an unweighted neighbor mean; composition-only retains posterior weighting but fixes the coefficient to one-half. Fixed-prior uses the original posterior without estimating π . All four rules use exactly R2’s support-only five-shot fallback. The nine uncentered mix/penalty controls use mixture coefficients in $\{0, 0.25, 0.5\}$ and penalties in $\{0.01, 0.1, 1\}$ for one shot, with the same five-shot fallback. Centered controls use their stated penalty for both support counts. LR1 and LR10 retain the support-only definition above.

Evaluation and findings. Each transfer direction uses five seeds with 100 tasks per support count and 15 queries per class. The two gallery conditions share task indices, giving 2,000 distinct sampled tasks and 4,000 task-condition evaluations. Three thousand paired bootstrap replicates preserve equal seed representation and average gallery conditions within task for the one-shot macro contrast. Intervals describe task sampling conditional on the fixed pools.

Table 6: One-shot paired accuracy differences in development transfer (percentage points). Each task is averaged across its two gallery conditions before resampling. The final column compares calibrated relevance with the same-mass control.

Training $\rightarrow$ target	Source-selected control	Δ	95% interval	Δ_{mass}
DTD $\rightarrow$ EuroSAT	CS $_{\ell_2}$, $\lambda = 0.1$	+0.48	[+0.30, +0.67]	0.00
EuroSAT $\rightarrow$ DTD	Mix 0.5, $\lambda = 0.01$	-1.45	[-1.72, -1.17]	0.00

The positive DTD-to-EuroSAT contrast in Table 6 does not establish better gallery composition: the same-mass control makes identical predictions in every evaluated condition. Reverse transfer

Table 7: Complete development-transfer accuracy (%). Each target uses the calibrator learned on the other domain. M and X denote matched and fully mismatched galleries. Each column contains 500 paired tasks. Five-shot fallback is specified in the text; duplicate rows reflect identical predictions rather than independent evidence.

Method	DTD				EuroSAT			
	1-shot		5-shot		1-shot		5-shot	
	M	X	M	X	M	X	M	X
R2	78.81	75.12	90.57	90.57	73.67	65.87	86.82	86.82
$CS_{\ell_2}, \lambda = 0.01$	78.90	74.86	90.27	90.27	73.35	67.93	89.37	89.37
$CS_{\ell_2}, \lambda = 0.1$	78.89	74.86	90.36	90.36	73.26	67.92	89.39	89.39
$CS_{\ell_2}, \lambda = 1$	78.87	74.73	90.56	90.56	72.57	67.66	88.24	88.24
Mix 0, $\lambda = 0.01$	75.49	75.49	90.57	90.57	68.56	68.56	86.82	86.82
Mix 0, $\lambda = 0.1$	75.55	75.55	90.57	90.57	68.46	68.46	86.82	86.82
Mix 0, $\lambda = 1$	75.61	75.61	90.57	90.57	67.65	67.65	86.82	86.82
Mix 0.25, $\lambda = 0.01$	77.70	75.81	90.57	90.57	71.67	67.79	86.82	86.82
Mix 0.25, $\lambda = 0.1$	77.66	75.79	90.57	90.57	71.49	67.86	86.82	86.82
Mix 0.25, $\lambda = 1$	77.46	75.74	90.57	90.57	69.99	67.38	86.82	86.82
Mix 0.5, $\lambda = 0.01$	78.88	75.13	90.57	90.57	74.14	65.47	86.82	86.82
Mix 0.5, $\lambda = 0.1$	78.81	75.12	90.57	90.57	73.67	65.87	86.82	86.82
Mix 0.5, $\lambda = 1$	78.28	74.91	90.57	90.57	71.77	65.54	86.82	86.82
LR1	75.58	75.58	90.47	90.47	67.62	67.62	85.12	85.12
LR10	75.41	75.41	90.68	90.68	68.24	68.24	88.17	88.17
Calibrated	75.55	75.55	90.57	90.57	73.67	68.46	86.82	86.82
Same mass	75.55	75.55	90.57	90.57	73.67	68.46	86.82	86.82
Composition only	78.32	74.99	90.57	90.57	73.67	65.39	86.82	86.82
Fixed prior	75.76	75.57	90.57	90.57	73.37	67.65	86.82	86.82

loses against its source-selected control. The five-shot fallback also trails the centered control on EuroSAT. The prespecified requirement of positive one-shot intervals against the source-selected and strong controls in both directions, plus no cell loss exceeding 0.5 percentage points, is therefore not satisfied.

On matched EuroSAT, the mean estimated membership prior is effectively 1 despite a true fraction of 0.5. On matched DTD it is effectively 0 despite a mean true fraction of 0.1064. Fully mismatched galleries also produce priors near zero. In this setup the correction behaves mainly as a domain-dependent switch for gallery influence. Independent numerical maximization on four retained tasks agrees with the implemented estimates within 10^{-5} . These findings are compatible with score miscalibration or class-conditional shift; they neither isolate a single cause nor rule out other learned relevance models. Exact five-shot fallback, task identities, source-fold separation and prediction-to-accuracy consistency were verified. No new target dataset is evaluated by this experiment.